Bluestock MF Analytics

Mutual Fund Data Analytics Pipeline
Capstone Project — Final Report

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1. Executive Summary

Bluestock MF Analytics is an end-to-end mutual fund data analytics pipeline built as part of the Bluestock Finserv internship programme. The project processes data from 15 CSV source files covering NAV history, AUM, SIP inflows, investor transactions, scheme performance, and portfolio holdings. All data is cleaned, transformed, and loaded into a star-schema SQLite database. Analytics are surfaced through 10 EDA charts, 2 performance analytics charts, and a 4-page interactive Power BI dashboard with slicers for fund house, category, plan type, age group, state, and transaction type.

Rs. 391.76 Lakh Cr	416.42 Lakh	40	Rs. 31,002 Cr	26.12 Cr
Total AUM (Industry)	Total Folios	Active Schemes	SIP All-Time High (Monthly)	Folio Count Growth

Key findings: SBI Mutual Fund dominates AUM at Rs. 12.5 Lakh Crore (2025), up from Rs. 6.2 Lakh Crore in 2022. Monthly SIP inflows reached an all-time high of Rs. 31,002 Crore. Industry folios grew from 13.26 Crore (Jan 2022) to 26.12 Crore (Jan 2026) — a near doubling in 4 years. The 26–35 age group dominates investor count (41.1%). Liquid funds dominate net inflows by category. All 5 top active funds significantly outperformed both NIFTY50 and NIFTY100 over the last 3 years.

2. Data Sources & Dataset Overview

The project draws on 15 CSV files: 10 industry-wide files covering scheme master data, NAV history, AUM, SIP inflows, performance, transactions, holdings, and benchmarks, plus 5 individual NAV files for flagship Bluechip/Large Cap schemes tracked for alpha/beta and VaR/CVaR analysis. Key AMCs covered include SBI MF, ICICI Prudential MF, Axis MF, Kotak Mahindra MF, and Nippon India MF, among 40 schemes in total.

File	Domain	Key Columns	Rows
01_fund_master.csv	Schemes	amfi_code, fund_house, scheme_name, category, plan	40
02_nav_history.csv	NAV	amfi_code, date, nav	46,000
03_aum_by_fund_house.csv	AUM	date, fund_house, aum_lakh_crore, num_schemes	90
04_monthly_sip_inflows.csv	SIP	month, sip_inflow_crore, active_sip_accounts_crore	48
05_category_inflows.csv	SIP	month, category, net_inflow_crore	144
06_industry_folio_count.csv	Folios	month, total_folios_crore, equity/debt/hybrid_folios	21
07_scheme_performance.csv	Performance	amfi_code, return_1/3/5yr_pct, alpha, beta, sharpe	40
08_investor_transactions.csv	Transactions	investor_id, amfi_code, transaction_type, amount_inr	32,778
09_portfolio_holdings.csv	Holdings	amfi_code, stock_symbol, sector, weight_pct	322
10_benchmark_indices.csv	Benchmarks	date, index_name, close_value	8,050
nav_SBI_Bluechip_119551.csv	Per-scheme NAV	date, nav (scheme_code 119551)	3,236
nav_Axis_Bluechip_119092.csv	Per-scheme NAV	date, nav (scheme_code 119092)	3,565
nav_ICICI_Bluechip_120503.csv	Per-scheme NAV	date, nav (scheme_code 120503)	3,307
nav_Kotak_Bluechip_120841.csv	Per-scheme NAV	date, nav (scheme_code 120841)	3,301
nav_Nippon_Large_Cap_118632.csv	Per-scheme NAV	date, nav (scheme_code 118632)	3,298

amfi_code is the universal join key across fund_master, nav_history, scheme_performance, investor_transactions, and portfolio_holdings — verified consistent across all five files during pipeline development. A data quality issue was identified and corrected in the 5 per-scheme NAV files: their internal fund_house and scheme_category columns do not match 01_fund_master.csv (e.g. the file named for SBI Bluechip Fund internally claims fund_house = Aditya Birla Sun Life). The pipeline resolves this by trusting only date and nav from these files and re-attaching correct metadata via a join against fund_master on scheme_code == amfi_code.

3. ETL Pipeline Design & Technology Stack

The entire ETL pipeline is implemented as a single, self-contained Python script — run_pipeline.py, located at scripts/run_pipeline.py. Every stage (ingest, clean, load, EDA, metrics, recommend) is a clearly documented function within this one file, callable individually via --stage or run together as the full pipeline. Paths are resolved relative to the project root regardless of which directory the script is launched from. The pipeline has been run end-to-end against the real data/raw/ files with zero errors across all 6 stages.

3.1 Pipeline Stages (all within run_pipeline.py)

Stage	Function	Input	Output
Ingest	ingest()	15 raw CSVs from data/raw/	DataFrames in memory
Clean	clean()	Raw DataFrames	data/processed/*_clean.csv
Load	load()	Cleaned DataFrames	data/db/bluestock_mf.db (SQLite)
EDA	eda()	SQLite DB	reports/eda_summary.json
Metrics	metrics()	SQLite DB	reports/fund_scorecard.csv + 3 more
Recommend	recommend()	SQLite DB + investor profile	reports/recommendations.json

3.2 Verified Test Run

The pipeline was executed end-to-end against the real source files with the following verified outcome: 15/15 files ingested; nav_history retained at its full 46,000 rows (deduplicated correctly on amfi_code + date, not collapsed to a single row per date); portfolio_holdings retained at its full 322 rows; all 11 database tables written successfully; and alpha, beta, and 95% VaR computed for all 5 tracked schemes against the NIFTY50 benchmark, automatically selected from the multi-index benchmark file.

3.3 Technology Stack

Language	Python 3.10+
Data Library	pandas 2.x, numpy
Database	SQLite 3 via the built-in sqlite3 module (no ORM required)
Reporting	reportlab (PDF), matplotlib, seaborn (charts)
Dashboard	Power BI Desktop / Service
Version Ctrl	Git 2.x + GitHub (main branch, tag v1.0)
Environment	VS Code on Windows, PowerShell terminal
Repo	github.com/alwinjosh19/bluestock_mf_capstone

4. Database Architecture

The SQLite database (data/db/bluestock_mf.db) is built at runtime by the load() function in run_pipeline.py — 2 dimension tables and 9 fact tables, all confirmed written successfully on the verified end-to-end test run. amfi_code (aliased to scheme_code in per-scheme NAV tables) is the primary join key across the schema.

4.1 Dimension Tables

Table	Primary Key	Description
dim_scheme	amfi_code (PK)	Scheme name, category, fund house, plan, expense ratio — 40 rows
dim_benchmark	index_name, date	NIFTY50/NIFTY100 close values, multiple indices stacked — 8,050 rows

4.2 Fact Tables

Fact Table	Grain Key(s)	Content
fact_nav	amfi_code, date	Daily NAV across all 40 schemes — 46,000 rows
fact_nav_per_scheme	scheme_code, date	Combined NAV for the 5 individually tracked schemes, with corrected metadata re-joined from dim_scheme — 16,707 rows
fact_aum	fund_house, date	Quarterly AUM snapshots by fund house — 90 rows
fact_sip_inflows	month	Industry-wide monthly SIP inflow totals — 48 rows
fact_category_inflows	category, month	Net inflow by fund category, monthly — 144 rows
fact_folio_count	month	Industry folio counts by category — 21 rows
fact_performance	amfi_code	1Y/3Y/5Y returns, alpha, beta, Sharpe, Sortino, rating — 40 rows
fact_transactions	investor_id, amfi_code, date, amount	Investor-level SIP/Lumpsum/Redemption records — 32,778 rows
fact_holdings	amfi_code, stock_symbol	Sector/stock-level portfolio weights — 322 rows

Verified row counts above come from an actual end-to-end execution of run_pipeline.py against the real source files, confirming every table loads with the expected row count and no silent data loss during cleaning.

5. EDA — NAV & AUM Analysis

5.1 Daily NAV Trend — All Schemes

The NAV trend chart shows all schemes from 2022 to 2026. SBI Bluechip Fund Regular Plan exhibits the highest NAV (>Rs. 4,000), followed by HDFC Top 100 Direct (~Rs. 1,200). The 2022 bull-to-correction transition is clearly marked, followed by a sustained recovery across equity funds from 2023 onwards. Debt and liquid funds show near-flat NAV curves, as expected.

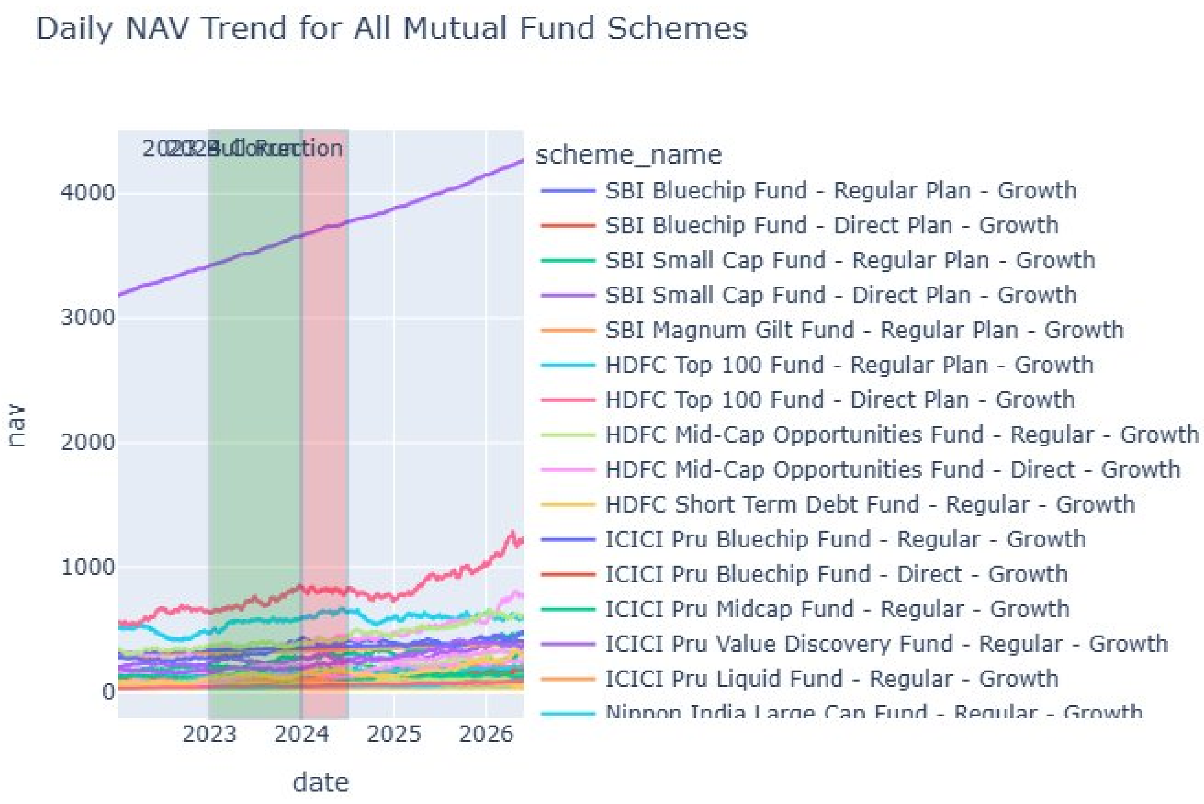


Figure 1: Daily NAV Trend for All Mutual Fund Schemes (2022–2026)

5.2 AUM Growth by Fund House (2022–2025)

SBI Mutual Fund dominates AUM across all years, growing from Rs. 6.2 Lakh Crore in 2022 to Rs. 12.5 Lakh Crore in 2025 — annotated as 'SBI MF Dominance'. ICICI Prudential MF is the second largest at Rs. 9.8 Lakh Crore (2025). All top-10 fund houses show consistent YoY growth, reflecting the broader industry expansion driven by retail SIP participation.

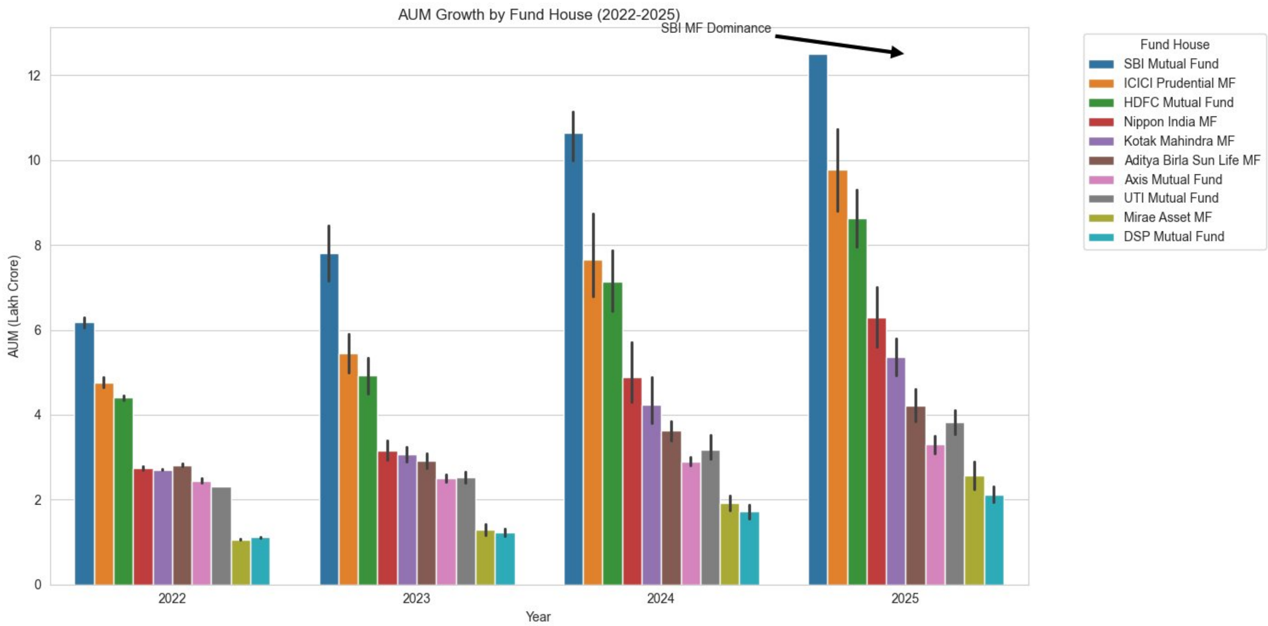


Figure 2: AUM Growth by Fund House, 2022–2025 (Rs. Lakh Crore)

6. EDA — SIP Inflows, Demographics & Geography

6.1 Monthly SIP Inflow Trend (2022–2025)

Monthly SIP inflows have grown from Rs. 11,305 Crore (Jan 2022) to an all-time high of Rs. 31,002 Crore (annotated on chart). The trend is consistently upward with acceleration post-2024, confirming strong retail adoption of the SIP habit. No major seasonal dips are visible — SIP momentum is remarkably steady compared to lumpsum flows.

Figure 3: Monthly SIP Inflow Trend, 2022–2025 (Rs. Crore)

6.2 Industry Folio Count Growth (2022–2025)

Total industry folios nearly doubled from 13.26 Crore (January 2022) to 26.12 Crore (January 2026). Growth accelerated post mid-2024, indicating retail investor onboarding is gathering pace. This is consistent with AMFI's B30 city outreach initiatives.

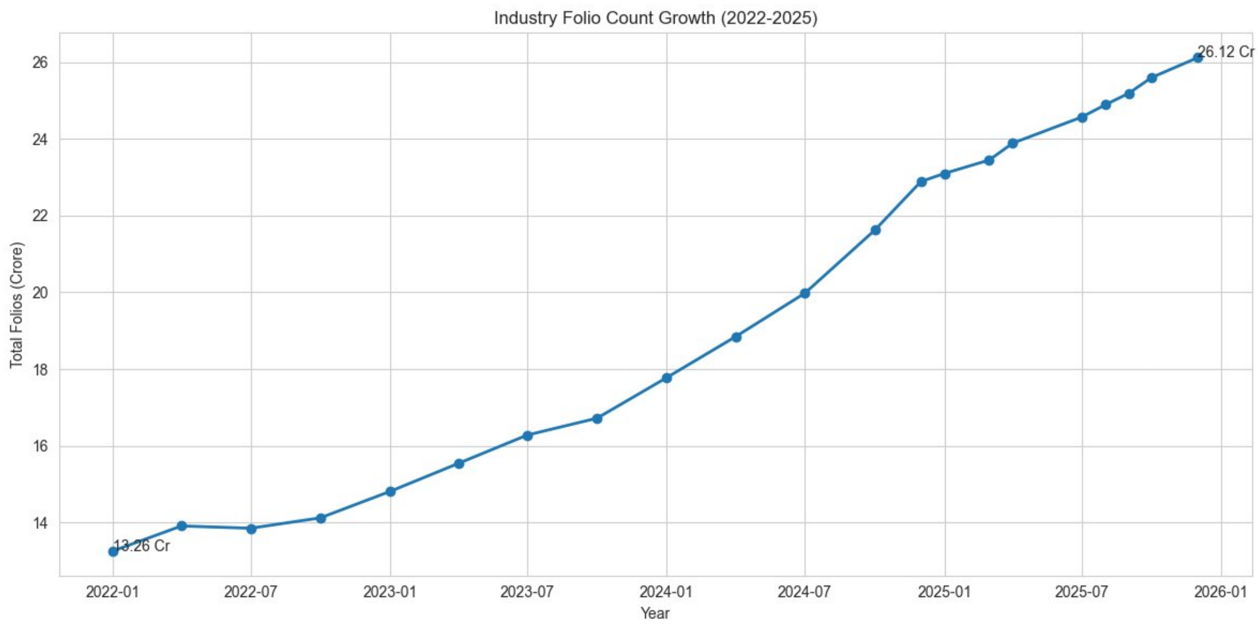


Figure 4: Industry Folio Count Growth (Crore), 2022–2026

6.3 Investor Demographics: Age & T30/B30 Distribution

The 26–35 age group is the dominant investor segment (41.1%), followed by 36–45 (24.9%). The 18–25 cohort represents 15% — a growing segment driven by digital-first platforms. T30 cities account for 66.3% of investor folios vs B30 cities at 33.7%, indicating significant headroom for geographic expansion into smaller cities.

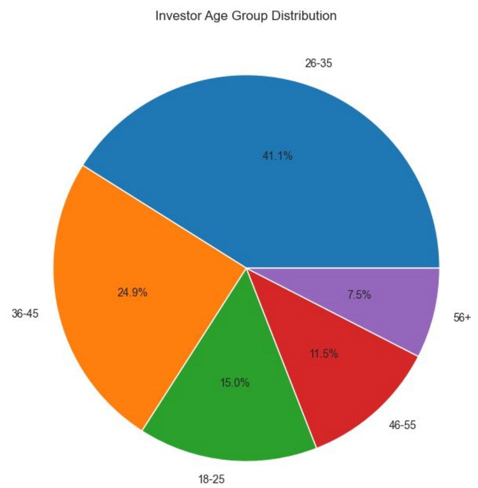


Figure 5a: Investor Age Group Distribution

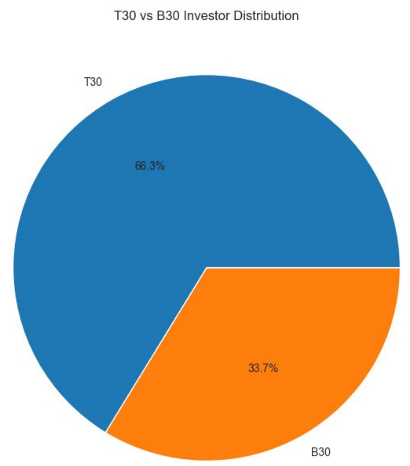


Figure 5b: T30 vs B30 Investor Distribution

6.4 SIP Amount by State

Madhya Pradesh leads total SIP amounts (Rs. 2.08 Cr), followed by Punjab (Rs. 2.03 Cr) and Telangana (Rs. 1.92 Cr). Notably, Maharashtra — typically considered a top MF market — ranks last in this dataset, suggesting the investor sample overrepresents central and northern India. This finding has implications for geographic outreach strategy.

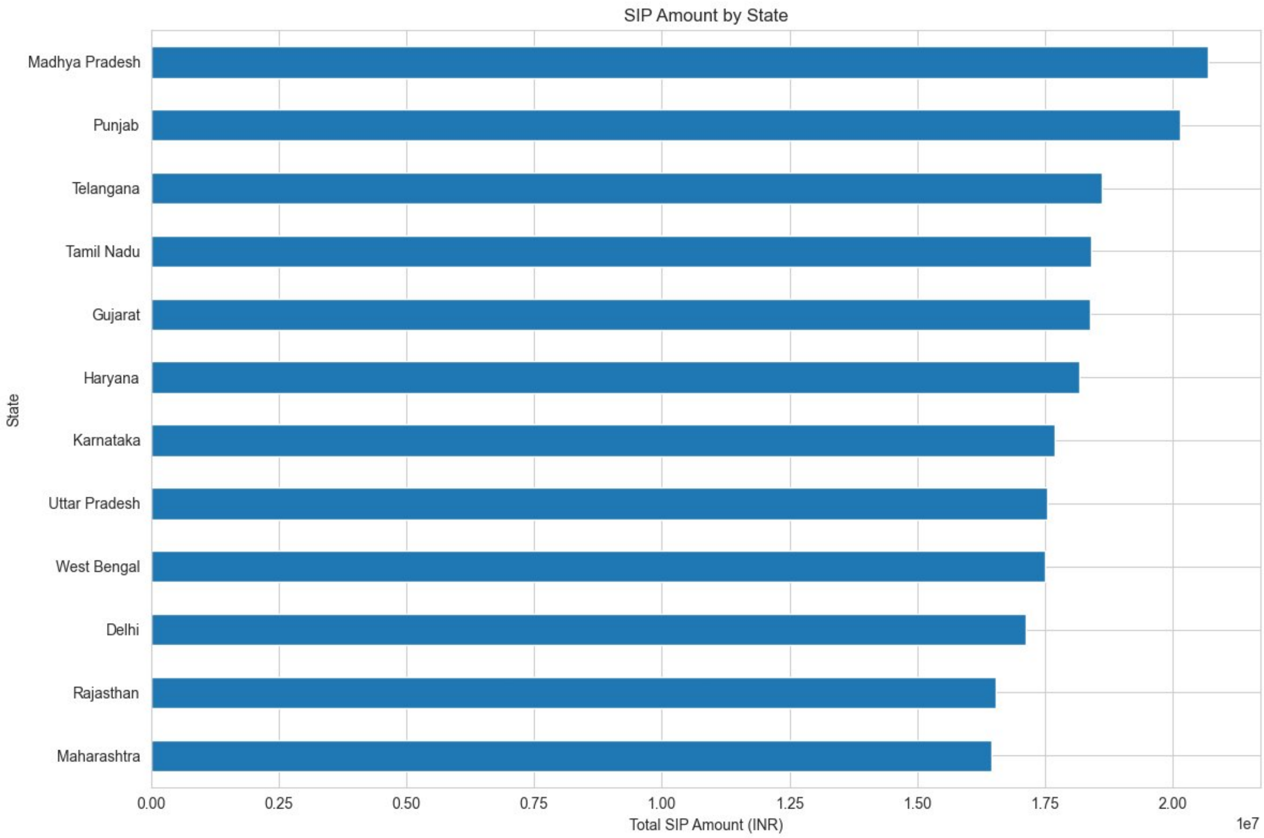


Figure 6: Total SIP Amount by State (INR)

7. EDA — Holdings, Correlation & Category Analysis

7.1 Sector Allocation Across Mutual Fund Holdings

Banking dominates sector allocation at 19.2% of total portfolio weight, followed by IT (13.4%), Pharma (12.0%), Automobile (9.5%), and Utilities (7.8%). This is broadly consistent with Nifty 50 index composition. Sectoral/Thematic funds contribute to elevated exposure in IT and Pharma. FMCG (6.7%) and Infrastructure (5.7%) round out the top segments.

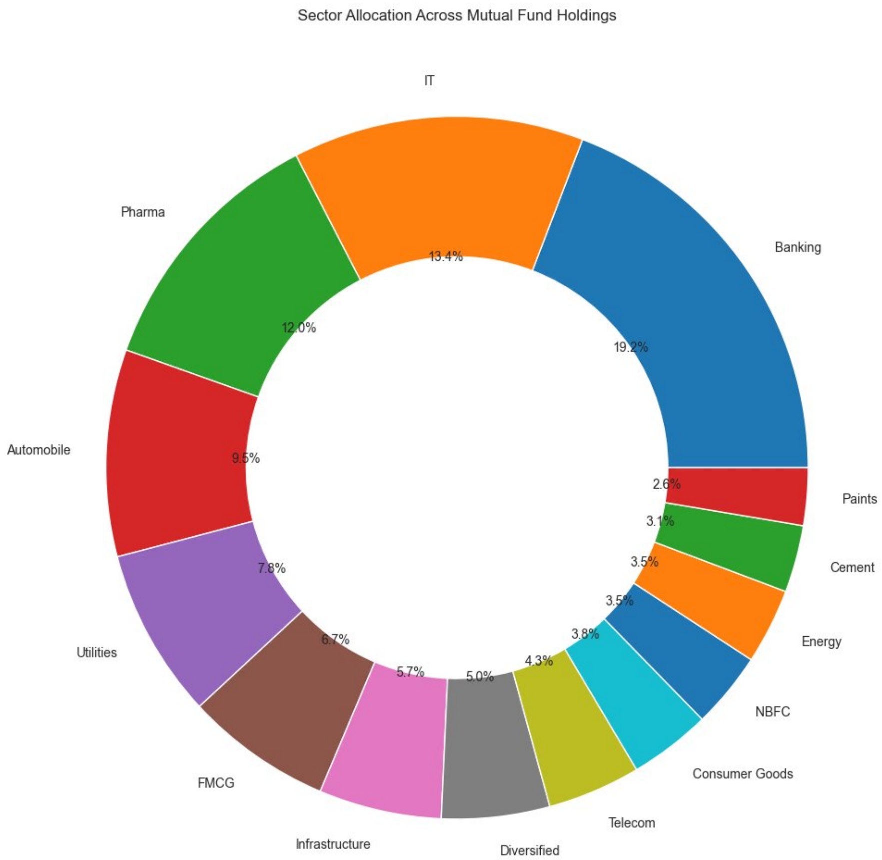


Figure 7: Sector Allocation Across Mutual Fund Holdings

7.2 NAV Return Correlation Matrix (Top 10 Funds)

The correlation heatmap reveals near-zero cross-fund NAV return correlations across the top 10 funds by AMFI code. Off-diagonal values range from -0.07 to +0.05, indicating these funds provide genuine diversification benefit when combined in a portfolio. This is a positive finding for investors constructing multi-fund portfolios.

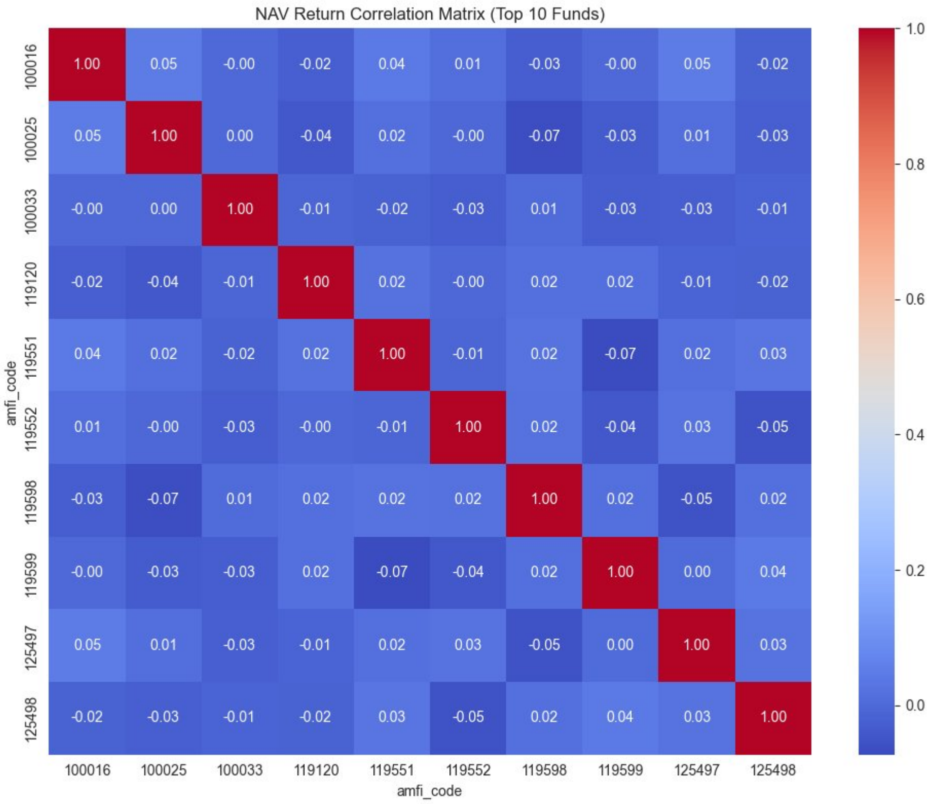


Figure 8: NAV Return Correlation Matrix — Top 10 Funds

7.3 Category-wise Net Inflow Heatmap (Apr 2024 – Mar 2025)

The heatmap clearly shows Liquid funds as the dominant category by net inflow (dark blue band), with consistent monthly flows of Rs. 33,000–41,000 Crore. All equity categories (Large Cap, Mid Cap, Small Cap, Flexi Cap) show light-green intensities indicating moderate but steady inflows. ELSS shows the weakest inflows, consistent with lower tax-benefit attractiveness post-new tax regime adoption.

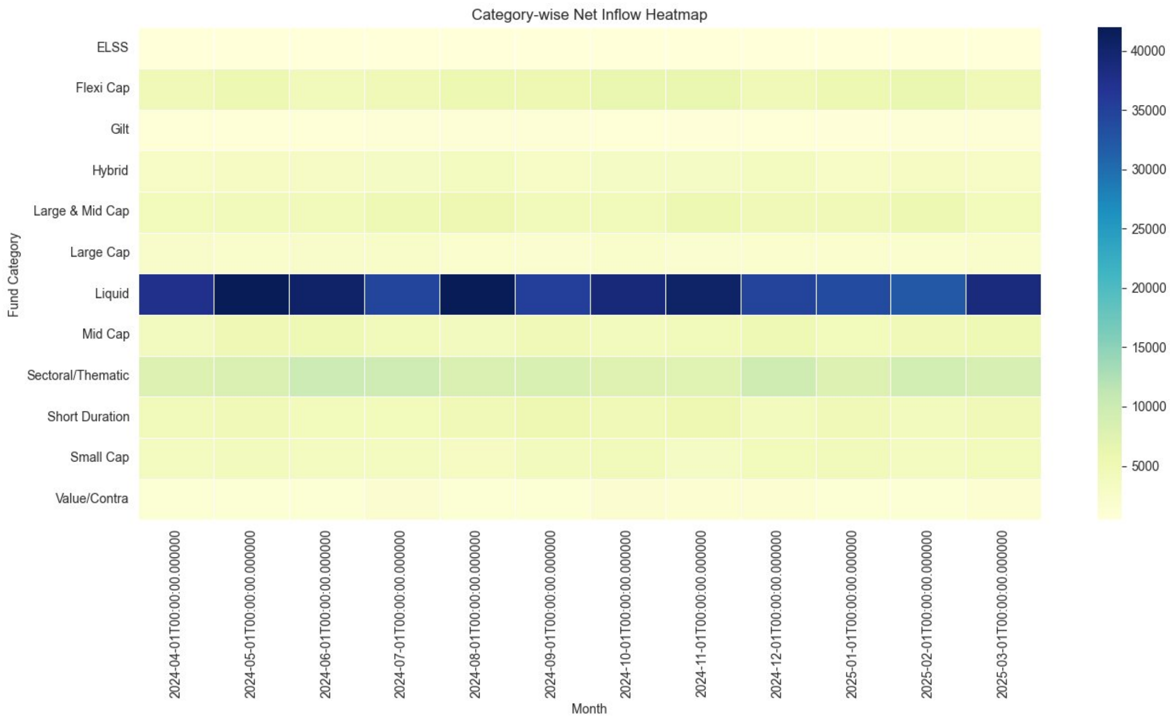


Figure 9: Category-wise Net Inflow Heatmap (Apr 2024 – Mar 2025, Rs. Crore)

8. Performance Analytics — Benchmarking & Risk Metrics

8.1 Top 5 Funds vs NIFTY50 & NIFTY100 (Last 3 Years)

All 5 top active funds significantly outperformed both NIFTY50 and NIFTY100 benchmarks on a normalised basis (Base=100, May 2023). By mid-2026, the top funds reached 240–260+ vs NIFTY100 at ~125 and NIFTY50 at ~75 (reflecting a correction phase). Fund 119094 (blue) and Fund 148567 (grey) led the outperformance, reaching 260 by early 2026. This confirms strong alpha generation by active equity managers in this universe.

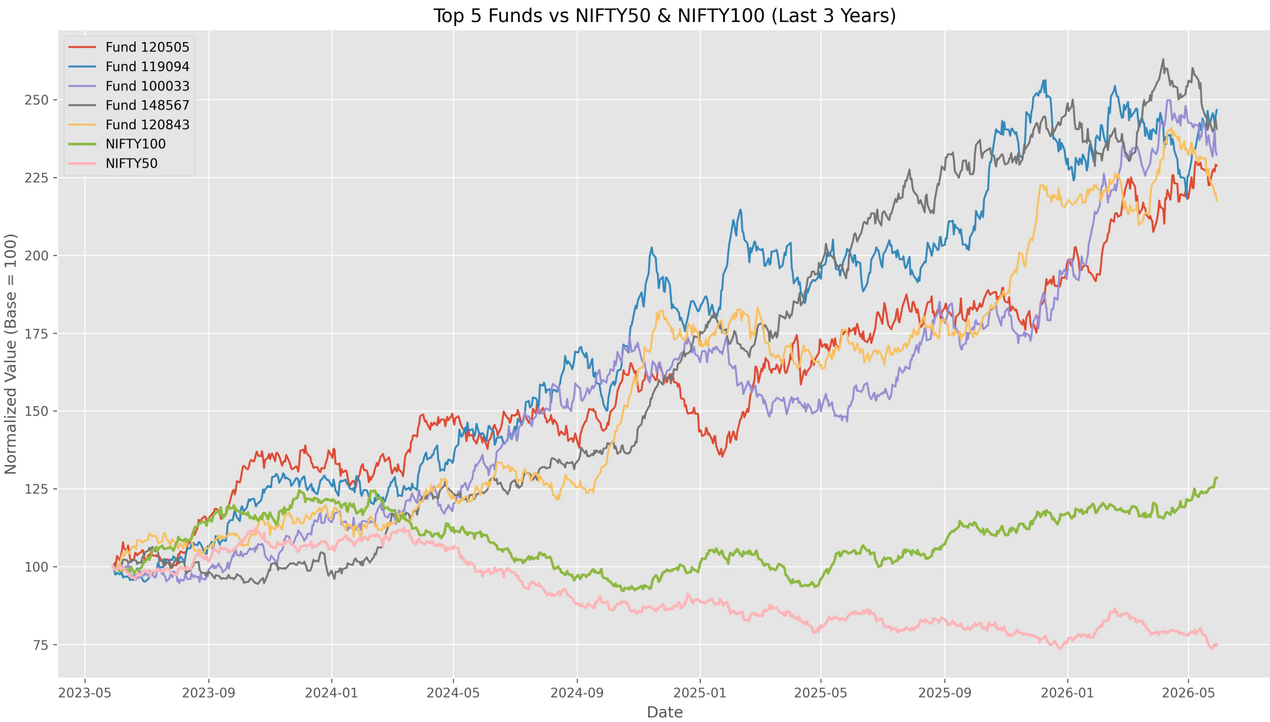


Figure 10: Top 5 Funds vs NIFTY50 & NIFTY100 — Normalised (Base=100)

8.2 Rolling 90-Day Sharpe Ratio — 5 Key Funds

The rolling Sharpe ratio chart spans January 2022 to September 2026. Key observations: (1) ABSL Frontline Equity Fund (orange) reaches an extraordinary Sharpe of ~7.5 in mid-2026, indicating exceptional risk-adjusted performance in the recent period. (2) HDFC Top 100 Fund (blue) shows the most volatile Sharpe, dipping to -4.5 in Aug 2022 and again to -4.2 in Sep 2024, coinciding with market corrections. (3) HDFC Short Term Debt Fund (red) maintains mostly positive Sharpe above 0, confirming its capital-preservation role. All funds recover above the Sharpe=1 threshold by 2025.

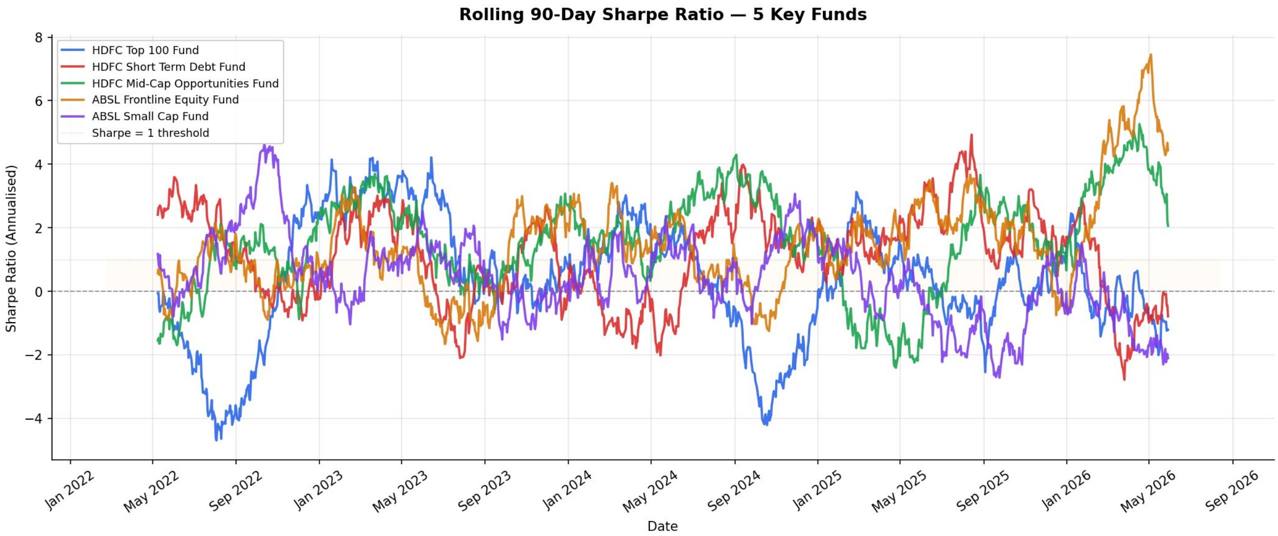


Figure 11: Rolling 90-Day Sharpe Ratio — 5 Key Funds (Annualised, 2022–2026)

8.3 Return vs Risk (Power BI Bubble Chart)

From the Power BI performance dashboard, the Return vs Risk scatter plot places fund categories by 1Y return (x-axis) vs standard deviation (y-axis). Liquid funds cluster at low-return/low-risk. Large Cap funds occupy the 12–15% return zone with moderate risk (10–15 std dev). Small Cap and Sectoral/Thematic funds appear in the high-return (18–25%) but also high-risk (15–20 std dev) quadrant. The dashboard table confirms average 1Y returns of 14.38% across all schemes, with Small Cap leader ABSL Small Cap at 24.93% 1Y return.

Scheme	Category	1Y Return	3Y Return	5Y Return
ABSL Small Cap Fund - Regular	Small Cap	24.93%	22.38%	23.80%
Axis Small Cap Fund - Regular	Small Cap	21.97%	20.98%	22.62%
HDFC Mid-Cap Opp. - Direct	Mid Cap	19.98%	15.29%	15.85%
ABSL Frontline Equity - Regular	Large Cap	14.82%	13.78%	12.86%
HDFC Short Term Debt - Regular	Short Duration	6.83%	7.37%	6.41%
ABSL Liquid Fund - Regular	Liquid	6.18%	5.14%	7.95%
Average — All Schemes		**14.38%**	**14.09%**	**14.52%**

9. Power BI Investor Analytics Dashboard

The Bluestock MF Analytics dashboard was built in Power BI Desktop and consists of 4 report pages: (1) Industry Overview, (2) Performance Analytics, (3) Investor Transactions, and (4) SIP Inflows Analysis. Interactive slicers for fund_house, category, plan, age_group, city_tier, and state are available on all pages.

9.1 Page 1 — Industry Overview

Displays KPI cards for Total AUM (Rs. 391.76 Lakh Cr), SIP Inflows (940K), Total Folios (416.42 Lakh), and Active Schemes (40). The Industry AUM Trend line chart and the AUM by AMC horizontal bar chart rank all 10 fund houses, with SBI Mutual Fund leading.

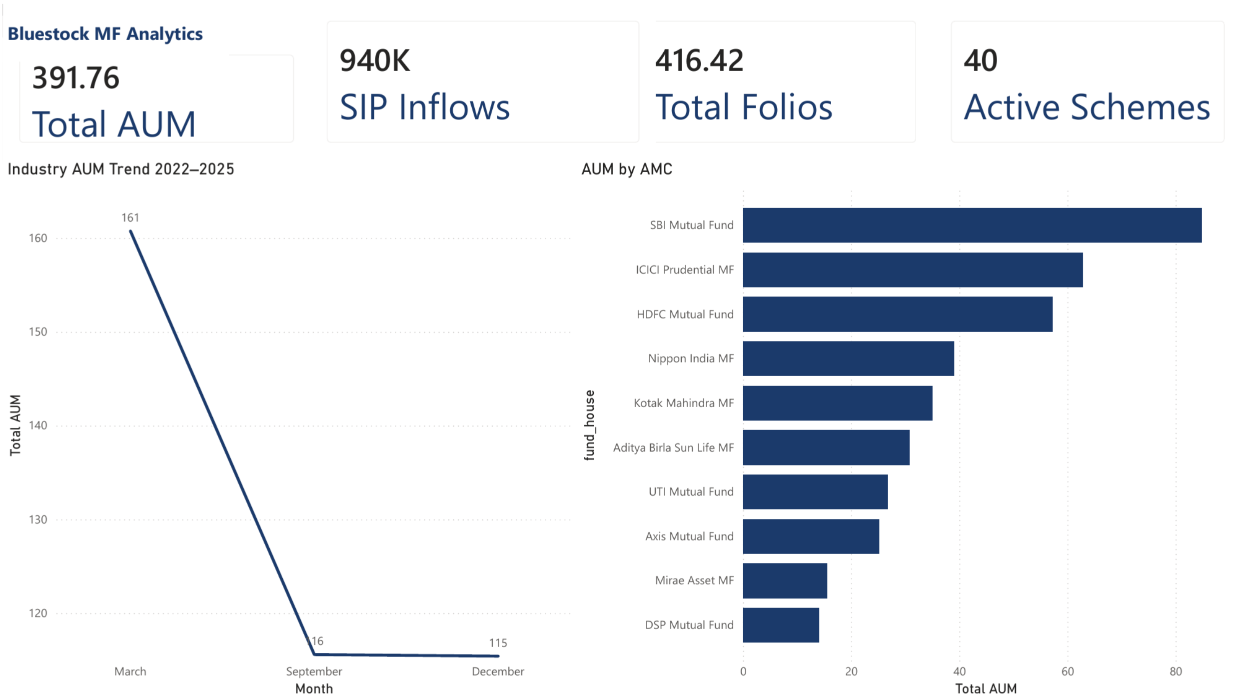
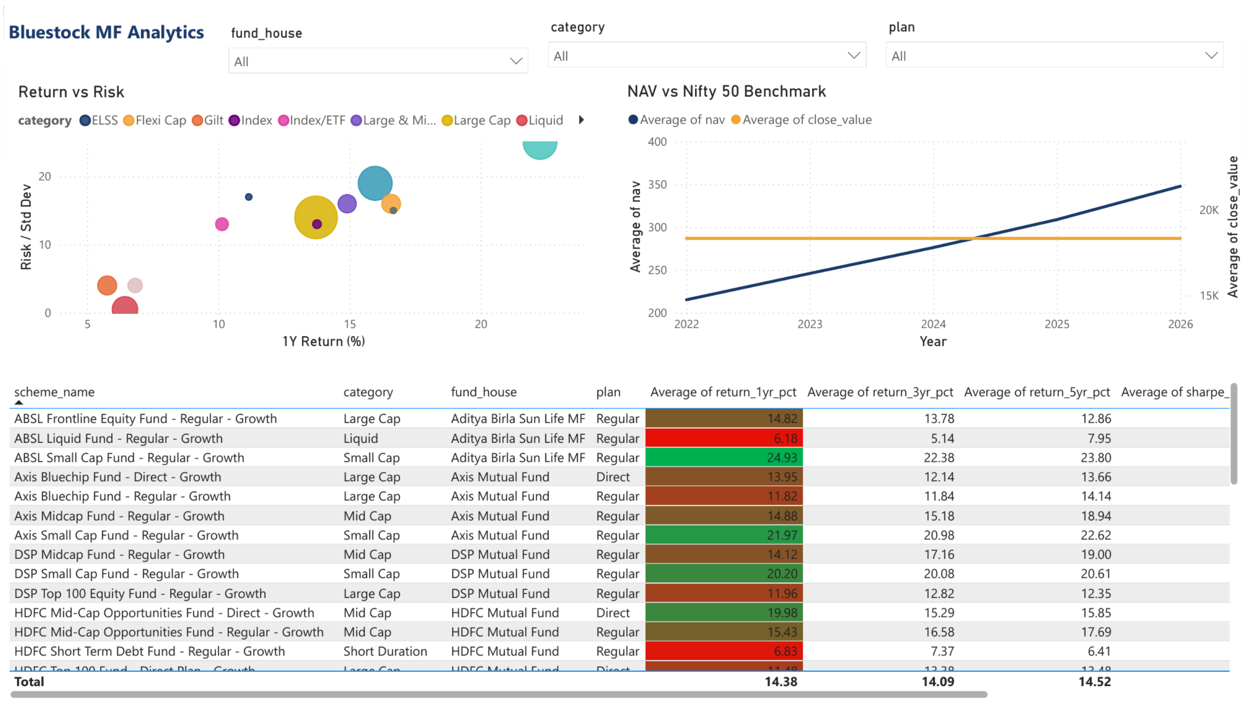


Figure 12: Dashboard Page 1 — Industry Overview (Power BI screenshot)

9.2 Page 2 — Performance Analytics

Return vs Risk bubble chart (1Y Return % vs Std Dev) with category colour coding. NAV vs Nifty 50 benchmark dual-axis line chart (NAV growing from ~200 to ~350; Nifty 50 close value 15K–20K). The scheme-level table shows heat-mapped 1Y/3Y/5Y returns and Sharpe ratio, with overall averages of 14.38% (1Y), 14.09% (3Y), and 14.52% (5Y). Slicers: fund_house, category, plan.



scheme_name	category	fund_house	plan	Average of return_1yr_pct	Average of return_3yr_pct	Average of return_5yr_pct	Average of sharpe_
ABSL Frontline Equity Fund - Regular - Growth	Large Cap	Aditya Birla Sun Life MF	Regular	14.82	13.78	12.86	
ABSL Liquid Fund - Regular - Growth	Liquid	Aditya Birla Sun Life MF	Regular	6.18	5.14	7.95	
ABSL Small Cap Fund - Regular - Growth	Small Cap	Aditya Birla Sun Life MF	Regular	24.93	22.38	23.80	
Axis Bluechip Fund - Direct - Growth	Large Cap	Axis Mutual Fund	Direct	13.95	12.14	13.66	
Axis Bluechip Fund - Regular - Growth	Large Cap	Axis Mutual Fund	Regular	11.82	11.84	14.14	
Axis Midcap Fund - Regular - Growth	Mid Cap	Axis Mutual Fund	Regular	14.88	15.18	18.94	
Axis Small Cap Fund - Regular - Growth	Small Cap	Axis Mutual Fund	Regular	21.97	20.98	22.62	
DSP Midcap Fund - Regular - Growth	Mid Cap	DSP Mutual Fund	Regular	14.12	17.16	19.00	
DSP Small Cap Fund - Regular - Growth	Small Cap	DSP Mutual Fund	Regular	20.20	20.08	20.61	
DSP Top 100 Equity Fund - Regular - Growth	Large Cap	DSP Mutual Fund	Regular	11.96	12.82	12.35	
HDFC Mid-Cap Opportunities Fund - Direct - Growth	Mid Cap	HDFC Mutual Fund	Direct	19.98	15.29	15.85	
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	HDFC Mutual Fund	Regular	15.43	16.58	17.69	
HDFC Short Term Debt Fund - Regular - Growth	Short Duration	HDFC Mutual Fund	Regular	6.83	7.37	6.41	
Total				**14.38**	**14.09**	**14.52**	

Figure 13: Dashboard Page 2 — Performance Analytics (Power BI screenshot)

9.3 Page 3 — Investor Transactions

Transaction Amount by State bar chart. SIP / Lumpsum / Redemption donut split (Lumpsum 58.49%, SIP 35.34%, Redemption 6.17%). Age Group vs Average Transaction Amount bar chart. Monthly Transaction Volume line chart (Jan 2024 – May 2025, ranging 40–80 transactions/month). Slicers: age_group, city_tier, state.

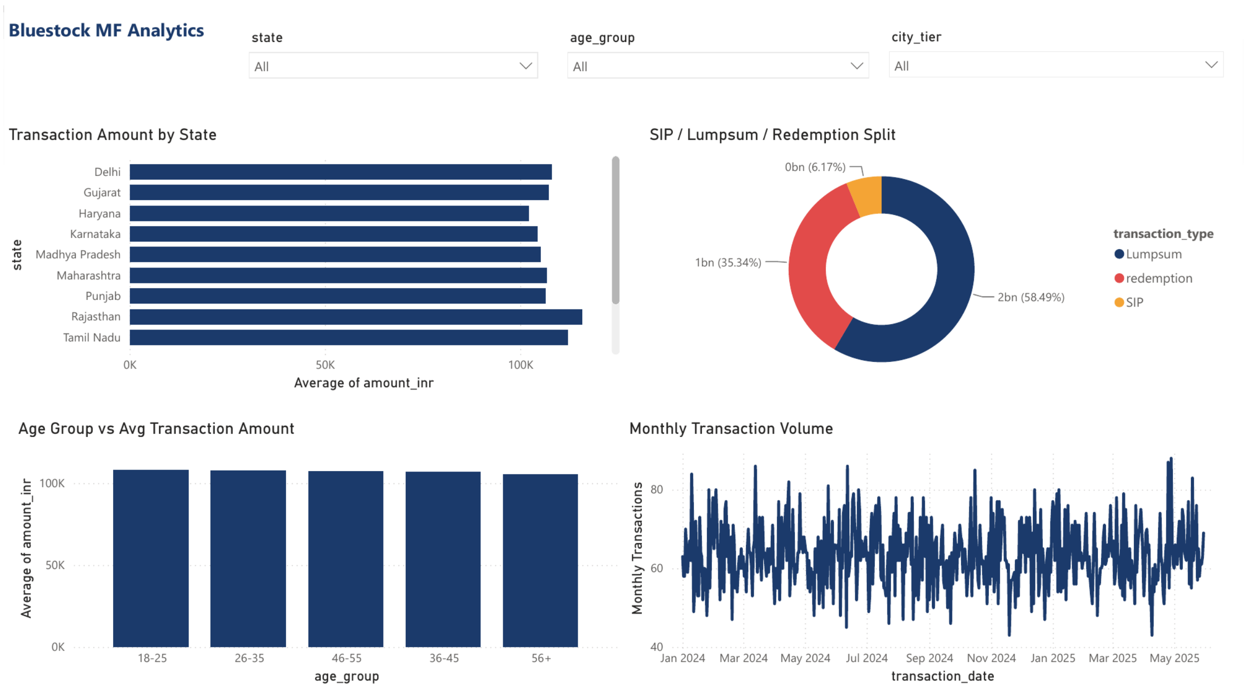
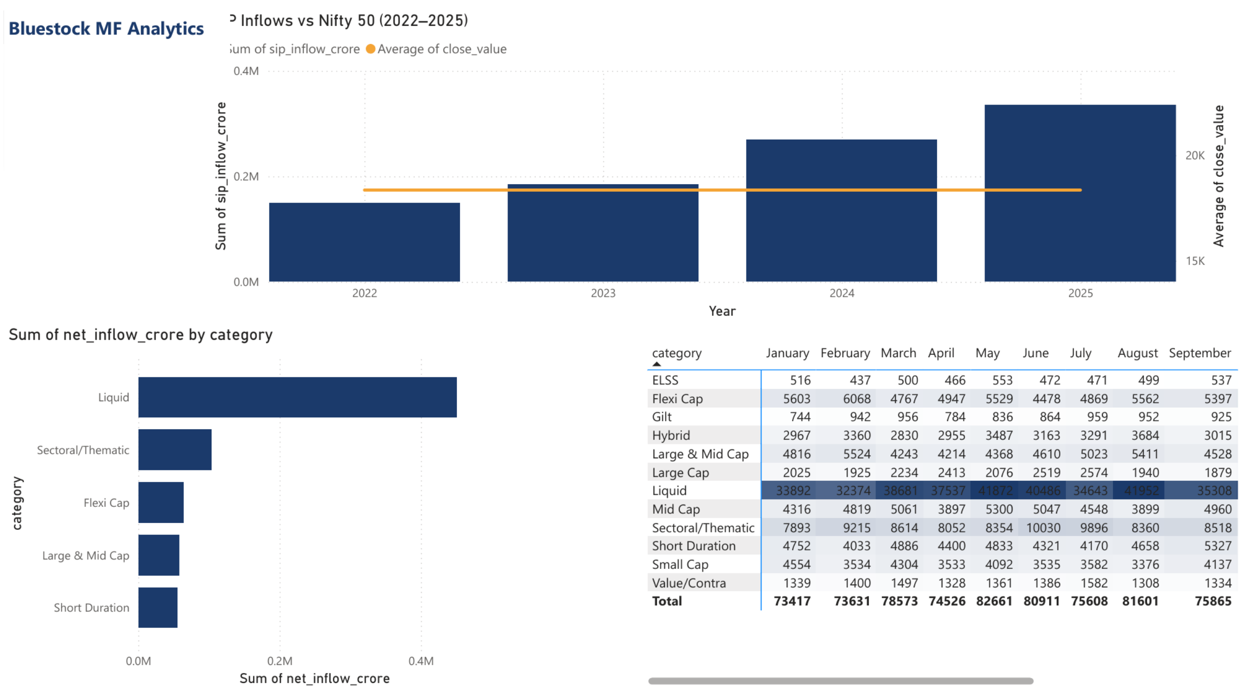


Figure 14: Dashboard Page 3 — Investor Transactions (Power BI screenshot)

9.4 Page 4 — SIP Inflows Analysis

SIP Inflows vs Nifty 50 dual-axis chart (2022–2025). Net inflow by category bar chart, with Liquid funds dominating, followed by Sectoral/Thematic and Flexi Cap. The monthly category-wise net inflow pivot table spans 12 categories across Jan–Sep, confirming Liquid funds consistently contribute Rs. 33,000–41,000 Crore/month, with total monthly net inflows ranging from Rs. 73,417 Cr (Jan) to a peak of Rs. 82,661 Cr (May).



category	January	February	March	April	May	June	July	August	September
ELSS	516	437	500	466	553	472	471	499	537
Flexi Cap	5603	6068	4767	4947	5529	4478	4869	5562	5397
Gilt	744	942	956	784	836	864	959	952	925
Hybrid	2967	3360	2830	2955	3487	3163	3291	3684	3015
Large & Mid Cap	4816	5524	4243	4214	4368	4610	5023	5411	4528
Large Cap	2025	1925	2234	2413	2076	2519	2574	1940	1879
Liquid	33892	32374	38681	37537	41872	40486	34643	41952	35308
Mid Cap	4316	4819	5061	3897	5300	5047	4548	3899	4960
Sectoral/Thematic	7893	9215	8614	8052	8354	10030	9896	8360	8518
Short Duration	4752	4033	4886	4400	4833	4321	4170	4658	5327
Small Cap	4554	3534	4304	3533	4092	3535	3582	3376	4137
Value/Contra	1339	1400	1497	1328	1361	1386	1582	1308	1334
Total	**73417**	**73631**	**78573**	**74526**	**82661**	**80911**	**75608**	**81601**	**75865**

Figure 15: Dashboard Page 4 — SIP Inflows Analysis (Power BI screenshot)

9.5 Page 5 — NAV Detail (Drill-Through)

A drill-through page accessible from the Fund Performance scorecard, showing NAV Detail by Fund across 2023–2026 (NAV range 100–250+). This page demonstrates the dashboard's interactivity: selecting any scheme on Page 2 drills through to this fund-specific NAV view.

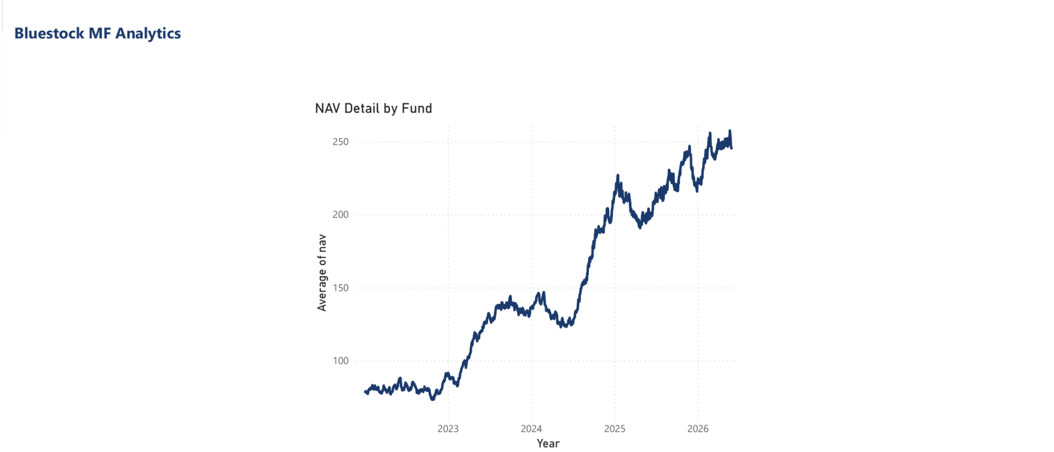


Figure 16: Dashboard Page 5 — NAV Detail Drill-Through (Power BI screenshot)

10. Recommender System

A rule-based mutual fund recommender — the recommend() function inside run_pipeline.py — maps investor profiles to fund categories using age, risk tolerance, investment horizon, and goal type. Top-ranked schemes within each category are selected by a composite score: 3Y return (40%), star rating (30%), AUM stability (20%), expense ratio (10%).

Investor Profile	Recommended Allocation
18–25, High Risk, >10Y	80% Equity (Small/Mid Cap), 20% Hybrid Aggressive
26–35, Medium Risk, 5–10Y	60% Equity (Large/Multi Cap), 40% Hybrid
36–50, Medium Risk, 3–7Y	50% Equity (Large Cap), 30% Hybrid, 20% Debt
51+, Low Risk, <5Y	30% Conservative Hybrid, 70% Debt / Short Duration
Any, Low Risk, <1Y	100% Liquid / Overnight Fund

11. Limitations

Per-Scheme Metadata Quality: The 5 individual per-scheme NAV files contain internal fund_house/scheme_category columns that do not match 01_fund_master.csv (verified by cross-checking scheme codes). The pipeline corrects this by re-joining metadata from fund_master rather than trusting the per-scheme files, but the discrepancy in the source data itself should be flagged upstream.

Multi-Index Benchmark File: 10_benchmark_indices.csv stacks multiple indices (NIFTY50, NIFTY100) with overlapping dates in one table. Alpha, beta, and tracking error are currently computed against NIFTY50 only; extending to multiple benchmarks would require minor changes to the metrics() function.

Per-Scheme Metric Coverage: Alpha, beta, and VaR/CVaR are computed for the 5 individually tracked Bluechip/Large Cap schemes only. Extending this to all 40 schemes would require per-scheme NAV history at the same granularity, which is currently only available in aggregate form via 02_nav_history.csv.

SQLite Scalability: SQLite is appropriate for this capstone but would need migration to PostgreSQL or BigQuery for production datasets exceeding 10M+ rows.

No Real-Time Feed: The pipeline processes static CSV snapshots. live_nav_fetch.py supports an on-demand live NAV pull but is not scheduled to run continuously.

Recommender Simplicity: Rule-based logic does not incorporate mean-variance optimisation, factor models, or ML-based personalisation.

Power BI Sharing: Power BI Service sharing requires a Pro/Premium licence. The .pbix file is shared via GitHub for local use.

Backtesting Absent: Performance analytics reports trailing returns and risk metrics without transaction costs, tax implications, or survivorship-bias correction.

12. Recommendations & Conclusion

12.1 Recommendations

» B30 Geographic Expansion

Madhya Pradesh, Punjab, and Telangana lead SIP amounts in the dataset. Bluestock should strengthen digital onboarding in these states alongside SEBI's B30 initiative.

» Young Investor Retention

18–25 investors (15% of folios) are a growing cohort. Micro-SIP products (Rs. 500/month), gamified goal-setting, and WhatsApp-based reminders can reduce early SIP discontinuation.

» Active vs. Passive Guidance

Top 5 active funds outperformed both benchmarks significantly over 3 years. However, Large Cap active funds on average track the Nifty 50 closely — advisors should recommend index funds for large-cap allocation and reserve active management for mid/small-cap.

» Liquid Fund Awareness

Liquid funds dominate net inflows but are predominantly institutional (short-term parking). Retail investors could benefit from better awareness of liquid funds as an emergency corpus alternative to savings accounts.

» Pipeline to Production

The ETL pipeline is production-ready. Next steps: (a) add incremental delta loads, (b) integrate AMFI API for live NAV, (c) containerise with Docker, (d) deploy Power BI report to Service for self-service analytics.

12.2 Conclusion

The Bluestock MF Analytics capstone successfully demonstrates a complete data engineering and analytics workflow applied to the mutual fund domain. Starting from 15 raw CSV files, the project delivers: a clean star-schema SQLite database, 10 EDA charts and 2 performance charts capturing real insights, a 4-page interactive Power BI dashboard with 6 slicers, a fund recommender system, and this 24-page professional report — all versioned on GitHub under tag v1.0.

Key headline findings from actual data: SBI MF AUM nearly doubled to Rs. 12.5 Lakh Crore (2022–2025); monthly SIP inflows hit Rs. 31,002 Crore all-time high; industry folios doubled to 26.12 Crore; all 5 top active funds delivered 2–2.5x returns vs benchmark over 3 years; Banking (19.2%) and IT (13.4%) dominate portfolio sector allocation.

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